

Vodafone and Idea merger

Vodafone India reaches with an agreement to acquire Idea to form India's largest operator with 377 million customer. <http://dgit.in/VodaIdea>



JPEG sized reduced

Google has developed a new JPEG algorithm that reduces file size by about 35 percent. <http://dgit.in/35jpeg>

Origins



Ken Thompson (sitting) and Dennis Ritchie working together at a PDP-11

a team of researchers led by himself and Dennis Ritchie, wanted to build a system “around which a fellowship could form”. One can safely say that they were successful.

Hello phreak

Come the 1970s, hacking temporarily took a new direction – telephone networks. While phone ‘phreaks’ had existed since the 1950s, it was AT&T’s implementation of fully automatic switches which used a tone dialing system which galvanised their popularity. One of the internal-use tones was of 2600 Hz, which caused a telephone switch to think the call had ended, leaving an open carrier line. This line could then be exploited to provide free long-distance, and international, calls. At that time, long-distance calls were quite expensive. Some of the most notable phone phreaks are Joe Engressia (a blind seven year old who had the perfect pitch needed for the hack), William ‘Bill’ Acker (who discovered that the recorder he owned could play the tone) and John ‘Cap’n Crunch’ Draper (who discovered, through his friendship with Engressia, that the whistle in Cap’n Crunch cereal boxes could also work).

Rise of the PC hacker

The 1980s essentially saw the first democratization of the computer – they weren’t out of reach anymore and the introduction of turnkey “personal” com-



Captain Crunch at Maker Faire Berlin (2015)

puters by Radio Shack, IBM, Apple and others had made computing accessible. With the advent of the ARPANET and modems connecting computers with each other being increasingly available, a hacker’s reach was further expanded.

Movies and novels that glorified hackers, or at least their abilities, increased in number around this time. War Games, a movie about an oblivious hacker almost starting World War III, is now held in cult status among hackers. Neuromancer, a hacking related novel released in 1984 is also credited by many as one of the main factors behind popularisation of the hacker among the general masses. Tron, Blade Runner, Cyberpunk and Terminator also rose in popularity around this time.

With this democratization came what can be termed as the ‘golden age for hacking’, although some of that gold might not have benefited everyone. The 1980s saw some of the first convictions for cyber crimes, including the first hacker convicted as a felon.

Good hacker bad hacker

In 1980, Ian Murphy (a.k.a. “Captain Zap”) along with three other hackers, got into

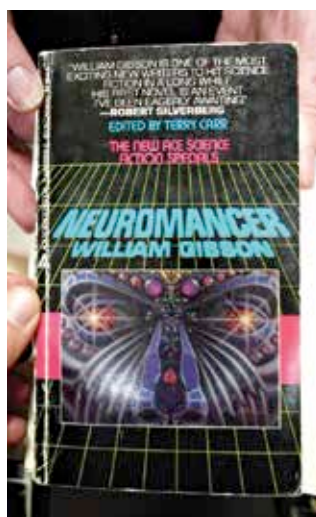
AT&T’s computers and altered their internal clocks. AT&T users suddenly started receiving late-night discounts in the afternoon, while others who waited until midnight to use the phone had to deal with massive bills. For his role in the crime, Murphy was greeted with 1,000 hours of community service and 2 1/2 years probation (considerably less than what hackers would receive today). He also inspired the movie ‘Sneakers’.

In another first, the FBI busted the Milwaukee-based 414s (named after the local area code), with the accusation of 60 computer break-ins ranging from the Memorial Sloan-Kettering Cancer Center to the Los Alamos National Laboratory. The 1980s saw more such criminal convictions of hackers who had, in some way, had a malicious effect on computer systems. The public were now more aware than ever of being hacked and the demand for security measures soared.

The wrong side of the law

In 1986, in response to all the hacking related felonies and convictions emerging across the country, the US Congress passed the Computer Fraud and Abuse act, the first legislation related to hacking. By then, the first major hacking groups had also been formed - the Legion of Doom in the United States and the Chaos Computer Club in Germany.

Pretty soon the two sides would face off in multiple instances. Operation Sundevil, which took place in 1990, was publicised as the largest crackdown on hackers. The crackdown took place in 15 cities and resulted in 3 arrests. As a response, the Electronic Frontier foundation was formed to fight for those accused of hacking in the very same year as Operation Sun Devil. Empowered by the



A first edition copy of the title ‘Neuromancer’